

DOCUMENT
DISCOVERY

PACK1

Activity 03 — Maintenance Service Simulation · Day 1

Expediente 1 — Descubrimiento · Actividad 03 · Día 1

Case / Caso	CASE 07 — Laboratorio Clínico Vida
Contracting mode / Modalidad	In-house IT staff
Date / Fecha	2026-08-25
Institution	INDEL — BTV en Desarrollo de Software · Module 3.6
Teacher / Docente	Ing. Javier Arturo García Mineros

Cantón Lourdes, Colón, La Libertad, El Salvador

Team and roles / Equipo y roles

MEMBER	ROLE	RESPONSIBILITIES IN THIS CASE (ENGLISH)
Lisette Menéndez	Service Manager	Coordinates client relations, manages maintenance window (22:00–01:00), approves scope, and signs delivery reports.
Jazmín Peña	Lead Technician	Executes diagnostics on network cables, switches, UPS, and server performance; prepares rollback plan.
Valeria Contreras	Documentation Officer	Maintains the bilingual document pack, logs maintenance activities, and prepares files for quality audit
Marjory Lima	Support & Communications	Handles communications with medical staff, sends maintenance notices, and provides user instructions.

Document 01 — System technical sheet / Lissett Menéndez

Documento 01 — Ficha técnica del sistema

ITEM	DESCRIPTION
System name and purpose	Clinical Laboratory Information Management System (LIMS) and Patient Results Portal.
Type (desktop app, web app, ERP, mobile, infrastructure)	Hybrid infrastructure: Client-Server desktop system for analysers and Web Application for patient portal.
Technology and version	Local LIMS database connected via serial/LAN interfaces to medical analysers.
Where the data is stored	Central On-Premise Server and Cloud Server for online patient portal results.
Number of workstations / devices	4 clinical analysers, 1 report printing station, 1 email PC, and 1 central server.
Number of users and profiles	Lab technicians (4), Administrative staff (2), External hospital users (3), Online patients.
Network and internet connection	Dedicated Ethernet Local Area Network (LAN) for analysers; High-speed Internet for portal and email.
Peripherals involved	Thermal report printers, 4 medical analysers, 1 failing UPS unit (beeping).
Existing backups	Daily local database snapshots, lacking recent audit and verification.
Existing documentation	Basic user manuals for analysers; no preventive maintenance log available.

Service hours of the client	24/7 continuous operation (Maintenance window restricted to 22:00–01:00).
Criticality (low / medium / high) and why	High: Operational failure risks patient diagnostic delays, data entry errors, and accreditation audit non-compliance.

Document 02 — Service request (ticket) / Jazmín Peña

Documento 02 — Solicitud de servicio

FIELD	CONTENT
Ticket number	TCK-2026-C07
Date and time received	2026-08-18 — 08:00 AM
Requested by (name and position)	Quality Manager — Laboratorio Clínico Vida
Channel (phone, e-mail, in person)	In-person internal escalation / Formal memo
Reported problem — in the client's words	"Two analysers lose server connection mid-batch, forcing manual typing. Web portal went down twice, and the UPS has been beeping for 2 weeks."
Impact on the business	High: Risk of manual diagnostic errors, delayed delivery to 3 hospitals, and audit non-compliance.
Urgency (low / medium / high)	High
Priority assigned and justification	High Priority: Immediate action needed due to risk of patient data corruption and 24/7 continuous operation requirement

Assigned to	In-House IT Support Team (VidaIT Maintenance Services)
Committed response date	2026-08-18 — 22:00 PM (Execution during the agreed maintenance window)

Document 03 — Scope statement / Marjory Lima

Documento 03 — Declaración de alcance

3.1 In scope

1. Diagnosis and replacement of the failing UPS battery/unit.
2. Physical and logical network check for the 4 lab analysers (switches and ethernet cables).
3. Performance check on the main server to resolve patient portal downtime.
4. Creation of a preventive maintenance log and schedule for the quality accreditation Audit

1. Diagnóstico y reemplazo de la batería o unidad UPS defectuosa.
2. Revisión física y lógica de la red de los 4 analizadores clínicos (switches y cableado).
3. Verificación de rendimiento del servidor para resolver las caídas del portal web.
4. Creación de una bitácora y programa de mantenimiento preventivo para la auditoría de acreditación.

3.2 Out of scope

1. Internal hardware or mechanical repairs on the medical analysers.
2. Maintenance of external hospital email servers.
3. Intervention of network infrastructure outside the laboratory facilities.

1. Reparación de hardware interno o fallas mecánicas propias de los analizadores médicos.
2. Mantenimiento de los servidores de correo externos de los hospitales.
3. Intervención de la infraestructura de red fuera de las instalaciones del laboratorio.

3.3 Assumptions and dependencies

1. The client provides total access to the server room during the 22:00–01:00 window.
2. Full system backups will be completed before starting any network or server work.
3. Technical support contacts from the analyser vendors are available on call.

1. El cliente otorga acceso completo a la sala de servidores durante la ventana de 22:00 a 01:00.
2. Se realizará un respaldo completo del sistema antes de iniciar cualquier trabajo en red o servidor.
3. Los contactos de soporte técnico de los proveedores de analizadores estarán disponibles ante cualquier eventualidad.

Client acknowledgement / Conformidad del cliente:

Name / Nombre: _____ Signature / Firma: _____
Date: _____

Document 04 — Discovery meeting minutes / Valeria Contreras

Documento 04 — Acta de reunión de descubrimiento

Date and time / Fecha y hora	August 18, 2026 — 10:00 AM
Place or channel / Lugar o medio	Main Meeting Room
Attendees / Asistentes	Lissette Menéndez (Service Manager), Jazmín Peña (Lead Tech), Quality Manager (Cliente)

4.1 Questions and answers

QUESTION ASKED (ENGLISH)	CLIENT'S ANSWER
What is the main problem you are facing right now?	The connection drops randomly, and the backup power battery is making loud noise.
How often does the web page stop working?	It went down two times this month, so patients could not check their results online.

What do you do when the machines disconnect?	We have to type all the test results into the system by hand, which takes a lot of time.
Is there any specific schedule we must follow to work?	Yes, you can only work at night between 10:00 PM and 1:00 AM because we never close.
Why is it urgent to fix this problem this week?	Because we have an upcoming quality audit and we need everything working perfectly.
How many samples do you process in a normal day?	We process around 300 medical samples every day for our patients and hospitals.
What happens with the computer files if the power goes out?	The UPS gives us a few minutes, but it beeps constantly and we fear losing data.

4.2 Agreements

Maintenance interventions will strictly take place between 22:00 PM and 01:00 AM. Full system database backups will be generated before restarting any network devices. The internal IT team is authorized to inspect and replace the failing UPS unit. A preventive maintenance schedule will be created for the upcoming quality audit.

Las intervenciones de mantenimiento se realizarán estrictamente entre las 22:00 PM y las 01:00 AM. Se generarán respaldos completos de la base de datos antes de reiniciar dispositivos de red. El equipo técnico interno queda autorizado para inspeccionar y reemplazar la unidad UPS defectuosa. Se creará un programa de mantenimiento preventivo enfocado en la próxima auditoría de calidad.

4.3 Pending information requested

Network map showing the IP addresses assigned to the 4 medical analysers. UPS model specifications and replacement battery availability. Analyser vendor support phone numbers in case interface assistance is needed.

Mapa de red con las direcciones IP asignadas a los 4 analizadores médicos. Especificaciones del modelo de la UPS y disponibilidad de baterías de repuesto.

Números de contacto del soporte del proveedor de los analizadores en caso de requerir asistencia de interfaz.

4.4 Preliminary hypothesis (to be verified)

What we think is happening:

- ✓ Analyser disconnections are caused by network congestion or damaged ethernet cables during peak data hours.
- ✓ The constant beeping of the UPS indicates a degraded internal battery failing to deliver stable voltage.
- ✓ Web portal downtime is related to high web traffic combined with server memory saturation.

What we must verify:

- ✓ Test network cables and switch ports connected to the 4 analysers.
- ✓ Measure battery health and voltage status of the UPS unit.
- ✓ Check server resource consumption (CPU and RAM) during peak hours.

Lo que creemos que está pasando:

La desconexión de los analizadores se debe a congestión de red o cables ethernet dañados durante horas de alta carga.

El pitido constante de la UPS indica una batería interna degradada que no entrega voltaje estable.

La caída del portal web está relacionada con alto tráfico de usuarios sumado a saturación de memoria del servidor.

Lo que debemos verificar:

Probar el cableado de red y los puertos del switch conectados a los 4 analizadores.

Medir el estado de la batería y el voltaje de la unidad UPS.

Revisar el consumo de recursos (CPU y RAM) del servidor en horas pico.

Questions

1. What exactly is the system?

It is a Clinical Laboratory Information Management System (LIMS) connected to 4 medical analysers, a central database server, a patient web portal, and a backup UPS.

2. Who uses it and when?

Lab technicians, administrative staff, external hospital users, and patients use it 24/7 in continuous operation.

3. What is failing, and what is only a symptom?

The root failures are a degraded UPS battery and network/server instability. The symptoms are the UPS beeping noise, analyser connection drops, and web portal crashes.

4. What is the cost to the client if nothing is done?

Manual typing errors in medical results, delayed diagnostic delivery to 3 hospitals, system crashes during power outages, and failing the quality accreditation audit.

5. What do we need to see or measure before promising anything?

We need to test the network cable continuity, measure switch port status, check server CPU/RAM usage during peak hours, and measure the UPS battery voltage.

6. What are we NOT going to do?

We are NOT going to repair the internal hardware of the medical analysers, modify external hospital servers, or perform maintenance outside the 22:00–01:00 window

Signatures / Firmas

The team declares that this file reflects the information obtained from the client and that no technical conclusion has been presented as a fact before being verified.

El equipo declara que este expediente refleja la información obtenida del cliente y que ninguna conclusión técnica se presenta como un hecho antes de ser verificada

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Service Manager

Lead Technician

Documentation Officer

